LABORATORY EVALUATION OF PRIMARY HEMOSTASIS

Tests **vascular** and **platelet response**

PLATELET EVALUATION

(1)QUALITATIVE TESTS: assesses platelet function/morphology

(2)QUANTITATIVE TESTS: assesses platelet count

QUALITATIVE PLATELET TESTS:

Tests for specific functions of platelet

(1)Adhesion

(2)Aggregation

(3)Clot retraction

(4)Factor assays

(1)PLATELET RETENTION TEST

- Principle: The **adhesiveness property of platelets** maybe assessed and measured by their ability to adhere on glass surfaces with beads
- Normal value: **75%-95%** of platelets should adhere

***Salzmann Method:** tests for the retention of platelets within the glass bead column

PLATELET ADHESION TEST (IN VITRO)

- **PLATELET RETENTION TEST/ GLASS BEAD RETENTION TEST**
  - Specimen: venous whole blood
  - Anticoagulant: **EDTA**
  - blood is passed thru a tube containing beads then platelets adheres to beads
  - To detect positive result: Requires 3 tubes
    - 1st tube control (platelet count)
    - 2nd tube get platelet count
    - 3rd tube get platelet count (must be lower because platelets adhere to beads)
    - Formula: 2nd tube – 3rd tube/ control tube X 100
  - Normal value: 75-95% platelets are lost
  - ABNORMAL PLATELET RETENTION TEST:
    - Decreased: **Bernard Soulier syndrome, Glanzmann's Thrombasthenia, vWD**, Chediak-Higashi, Myeloproliferative disorders, Uremia, Aspirin administration
    - Increased: Thrombotic disorders, Hyperlipedemia, Carcinoma, Oral contraceptives, Pregnancy
- **PLATELET ADHESIVENESS TEST/SALZMANN'S TEST**
  - PC 1: routine procedure, collection
  - PC 2: collected through glass bead collecting system

Formula: %PA: PC1-PC2/PC1 x 100

ABNORMAL PLATELET RETENTION TEST

DECREASED	INCREASED
✓ Bernard-soulier syndrome	✓ Thrombotic disorders
✓ Vwf DISEASE	✓ Hyperlipidemia
✓ Chediak-higashi	✓ Carcinoma
✓ Myeloproliferative disorders	✓ Oral contraceptives
✓ Uremia	✓ Pregnancy
✓ Aspirin admin.	

(2)TESTS FOR PLATELET AGGREGATION

A. PRP aggregometery

- ✓ Assess the ability of platelets to aggregate after addition of an aggregating agent
- ✓ Anticoagulant of choice: CITRATE
- ✓ Sample required: **PLATELET-RICH PLASMA**
- ✓ Initially the PRP is turbid due to the presence of platelets then by the addition of aggregating agents, *the platelet will then change its shape and later aggregate which decreases the turbidity of the specimen.*
- ✓ The decrease in the turbidity of the PRP is measured by the **platelet aggregometer**

- Aggregating agent: promote platelet aggregation
  - Abnormal aggregation with ACE/t: Glanzmann's Thrombasthenia
  - Abnormal aggregation with Ristocetin: von Willebrand's disease or Bernard Soulier Syndrome

AGGREGATING AGENT	NORMAL TRANSMITTANCE
ADP	Biphasic
Arachidonic acid	Biphasic
Collagen	Single curve
Epinephrine	Biphasic (last to be added)
Thrombin	Biphasic
Ristocetin	Broad (added first)
Serotonin	Low (weak aggregating agent)

B. PRP Whole Blood

- An equal amount of saline and WB
- Incubated at 37c for 5 minutes.
- Involves suspension of a pair of low voltage cartridge mounted disposabl electrodes.
- Based on electrical impedance

C. PLATELET LUMIAGGREGOMETRY

AGGREGATING AGENTS USED

- ✓ ADENOSINE DIPHOSPHATE (ADP)
- ✓ EPINEPHRINE
- ✓ THROMBIN
- ✓ ARACHIDONIC ACID
- ✓ RISTOCETIN
- ✓ COLLAGEN
- ✓ SEROTONIN

TEST CONSIDERATIONS

- ✓ No hemolysis
- ✓ Fasting: 8 hours
- ✓ pH: 6.5-8.5
- ✓ Temperature: 37°C
- ✓ Patient should not take any NSAIDS at least 7 hours before the test
- ✓ Performed within 3 hours

INTERFERING FACTORS:

- ✓ Platelet count: <100,000/uL
- ✓ Lipemia will interfere with testing
- ✓ 14 days before test, drugs that inhibit platelet aggregation are contraindicated (ex. aspirin, antihistamines, steroids, cocaine, anti-inflammatories, theophylline, and antibiotics)
- ✓ On the day of the test, avoid caffeine
- ✓ Avoid warfarin/coumadin for 2 weeks and heparin therapy for 1 week before testing

(3) TEST FOR CLOT RETRACTION

- Measure the entire function of platelets
- It also detects clotting factor deficiencies
- Depends on the normal levels of:
  - ✓ Fibrinogen
  - ✓ Calcium
  - ✓ ADP
  - ✓ Platelets

METHODS OF CLOT RETRACTION	
QUALITATIVE	**QUANTITATIVE**
✓ HIRSHBOECK METHOD -also known as :CASTOR OIL METHOD	✓ STEFANINI-DAMESHEK ✓ MCFARLANE METHOD

(4) PLATELET FACTOR ASSAYS

1.PF3 AVAILABILITY

- Platelet factor 3 is needed for the continuous activation of the clotting process
- This test uses PPP, PRP and activators such as kaolin or elagic acid to activate solely the platelets and the clotting of both specimens are determined.
- The clotting time of PRP incubated with clotting agent is considerably shorter than that of platelet poor plasma.

2. PF4 AND B-TG Assay

- Their presence in vivo indicates :
- RIA is the method used for this assay

QUANTITATIVE PLATELET TEST:

- part of coagulation workup/profile

- **PLATELET COUNT:** valuable in assessing bleeding disorders associated with thrombocytopenia, uremia, liver disease, or malignancies, and in monitoring the course of disease among patients with bone marrow failure

THROMBOCYTOPENIA

- Platelet count is BELOW the normal reference range
- diseases include:

  - **Marrow infiltrative processes** (ex:Leukemia,Myelofibrosis, Multiple Myeloma, Metastatic carcinoma)
  - **Aplasia** (ex: Aplastic anemia, Fanconi anemia, chemotherapy, and radiation
  - **Ineffective blood production** (ex: Vitamin B12 deficiency, Folic Acid deficiency, alcohol ingestion, myelodysplastic syndrome and PNH)
  - **Congenital disorders** (ex: Bernard Soulier Syndrome and May-Hegglin anomaly)

THROMBOCYTOSIS

- Platelet count is ABOVE the normal reference range
- diseases include:

1. Myelodysplastic syndromes

- ✓ Polycythemia vera
- ✓ Essential thrombocytosis

2. Reactive thrombocytosis

- ✓ After hemorrhage or surgical procedure
- ✓ Iron Deficiency Anemia
- ✓ Epinephrine administration
- ✓ Malignancies

Challenges in platelet count:

- Small size: 2-4um
- Adheres to foreign surfaces
- Easily disintegrate
- Hard to differentiate from debris
- Unevenly distributed in the blood since they tend to clump

3 WAYS TO QUANTIFY PLATELETS

1) PLATELET ESTIMATION USING PBS

INDIRECT PLATELET COUNT METHOD

- may detect PLATELET CLUMPING and CLOTS from poorly collected specimens
- NOT RELIABLE because the results depend upon the distribution of platelets and on the red cell count
- Platelets are counted in their relationship to RBCs on a fixed-stained smear

ADEQUATE PLATELET COUNT is reported when:

- 1 platelet per 10-40 RBC
- 3-10 platelets per 100 RBC (or in 1 OIO field)
- 5-20 platelets per 200 RBC
- 8-20 platelets per field (in 10 OIO fields)

INDIRECT PLATELET COUNT

1. FONIO'S METHOD

- Uses: 14% Magnesium Sulfate + 1 drop of blood + Wright stain

2. DAMESHEK METHOD

- Uses: Brilliant Cresyl Blue, sodium citrate, sucrose and formalin; then counterstained with Wright's stain
- Platelets/uL= $\frac{\text{plts counted x RBC count}}{1000}$
- Reference Range: 250-500 x 109/L or 250,000-500,000

INDIRECT PLATELET COUNT

3. MODIFIED INDIRECT METHOD

- Number of platelets are counted per field
- ADEQUATE PLATELET COUNT: 8-20 platelets/field
- Number of fields counted: 10

- Platelets/uL= $\frac{\text{plts counted in 10 fields x 20,000}}{10}$
- Reference range: 150-350 x 109/L or 150,000-350,000

PLATELET COUNT	ESTIMATION
0-49,000/uL	Markedly decreased
50,000-99,000/uL	Moderately decreased
100,000-149,000/uL	Slightly decreased
150,000-199,000/uL	Low normal
200,000-400,000/uL	Normal
401,000-599,000/uL	Slightly increased
600,000-800,000/uL	Moderately increased
>800,000/uL	Markedly increased

3 WAYS TO QUANTIFY PLATELETS

2) ABSOLUTE PLATELET COUNT USING HEMOCYTOMETER

DIRECT PLATELET COUNT METHOD

a. REES AND ECKER METHOD

- Also known as TONCANTIN'S METHOD
- Uses Rees-Ecker diluting fluid (storage temp: 0-4'C)
- Components of diluent: Distilled Water, Formalin Buffer and Brilliant Cresyl Blue Uses **RBC pipette (1:200)** or WBC pipette (1:20)

b. GUY AND LEAKE

3 WAYS TO QUANTIFY PLATELETS

2) ABSOLUTE PLATELET COUNT USING HEMOCYTOMETER

DIRECT PLATELET COUNT METHOD

c. BRECKER-CRONKITE

- Uses: Ammonium Oxalate
- Microscope: Phase Contrast Microscope

d. UNOPETTE SYSTEM

- Diluting fluid: Ammonium oxalate
- Dilution: 1:100
- It uses the entire RBC squares for counting platelet
- Platelet count formula: number of platelets X 1000

DIRECT PLATELET COUNT

REES AND ECKER	• Diluent: sodium citrate, formalin buffer, Brilliant Cresyl Blue, distilled water • Uses: LIGHT MICROSCOPE • Platelets appear as highly refractile round bodies • **Plts/uL= $\frac{\text{plts counted x 10 x 200}}{4}$**
GUY AND LEAKE	• Diluent: sodium oxalate, 40% formalin, and crystal violet • Uses: LIGHT MICROSCOPE • **Plts/uL= plts counted x 5 x 10 x 100**
BRECKER-CRONKITE	• Diluent: 1% Ammonium oxalate • Uses: PHASE CONTRAST MICROSCOPE • MOST ACCURATE METHOD • **Plts/uL= plts counted x area correction factor x 10 x 100**
UNOPETTE	• Diluent: 2% Ammonium oxalate • Uses: Bright field microscopy • It uses the entire RBC squares for counting platelet • Platelet count formula: number of platelets X 1000

Reference Range: 150-400 x 109/L or 150,000-400,000/uL

3 WAYS TO QUANTIFY PLATELETS

3. USE OF AUTOMATED COUNTING MACHINE

- The automated machine determines the platelet count depending on the average volume of the cell.
- RBC and Platelet are counted simultaneously by the machine and the machine sorted these 2 cells depending on their size or volume.
- Normal platelet volume is approximately **7 fL**
- If the volume falls within **2-20 fL** it will be registered by the machine as platelet

SOURCES OF ERROR

FALSELY INCREASED COUNT

- ✓ Fragments of leukocyte cytoplasm
- ✓ Inclusions from lysed RBCs
- ✓ Debris

FALSELY DECREASED COUNT

- ✓ Platelet satellitism
- ✓ Platelet clumping due to agglutinins
- ✓ Spontaneous platelet aggregation
- ✓ Clotting due to faulty blood collection and handling
- ✓ Delay in sampling

TEST OF PLATELET ADHESION

BLEEDING TIME

- Time required for a standardized wound to stop bleeding
- Standard skin wound:
- Tests the in **vivo function of platelets** and number
- Determines both congenital and acquired platelet disorder
- Factors affecting bleeding time:
  - ✓ Elasticity of tissue
  - ✓ Ability of the blood vessels to constrict and retract
  - ✓ Mechanical and chemical action of platelets in the formation of hemostatic plug

BLEEDING TIME

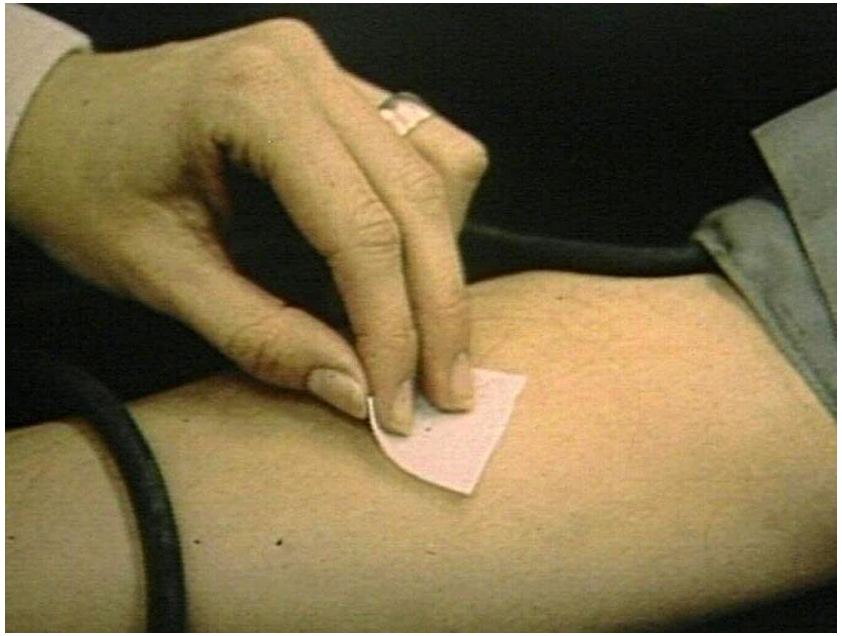
A. DUKE'S METHOD

- ✓ A standard wound is made on various site (ex. fingers toes, and earlobes)
- ✓ Site of choice: fingers
- ✓ Reference range: 2-6 minutes

B. IVY'S METHOD

- ✓ BEST SCREENING TEST available for platelet's role in hemostasis
- ✓ 2-3 **incisions** are made on the VOLAR SURFACE of the patients forearm with application of pressure usually around 40mmHg during the test
- ✓ Reference range: 2-9 minutes

MODIFIED IVY'S METHOD



BLEEDING TIME

C. MODIFIED SIMPLATE METHOD

- ✓ Modification of the Ivy's method
- ✓ It uses a **spring-loaded lancet** that will create exactly a standard skin wound on the chosen site
- ✓ Standard skin wound: **5mm x 1mm**

D. COPLY LALITCH METHOD

- ✓ Reference range: >3 minutes

E. ADELSON-CROSBY METHOD

- ✓ Same with Coply-Lalitch method

BLEEDING TIME

F. MACFARLANE's METHOD

- ✓ Same principle with Adelson-Crosby

G. ASPIRIN TOLERANCE TEST

- ✓ Assesses the effect of a standard dose of aspirin on the Duke's bleeding time

ABNORMAL BLEEDING TIME IS OBSERVED IN:

- ✓ Thrombocytopenia
- ✓ Hypofibrinogenemia
- ✓ vWF disorder
- ✓ Connective tissue disorders
- ✓ Scurvy
- ✓ Bernard Soulier Syndrome

TEST FOR VASCULAR RESPONSE (INTEGRITY)

- Measures the ability of the platelets to maintain capillary integrity.
- Results correlate with**: degree of thrombocytopenia** and **bleeding time**
- Clinical significance (+):
  - ✓ **Purpura**
  - ✓ **Dengue Fever**
  - ✓ **Scurvy**

1. TOURNIQUET TEST

- **Also known as: Capillary Fragility Test**, **Capillary Resistance Test**, **Rumple Leede CFT**, **Hess Test** and **Grocco-Frugoni Test**
- Uses **POSITIVE PRESSURE TECHNIQUE**
- Principle: By partially obstructing the flow of venous blood, the capillary pressure is increased. This will give rise **to intravasation of blood** which will be manifested in the form of small hemorrhage called *petechiae*
- It tests the stability of the small blood vessels to retain the red cell in their lumen under conditions of stress and trauma.
- Standard pressure: 40-60 mmHg
- Area where petechiae are counted: 1 inch

1. TOURNIQUET TEST

- Important factors:
  - ✓ Platelets
  - ✓ Vitamin C
  - ✓ Collagen

*All are important in the maintenance of normal capillary integrity or resistance.

- A. QUICK'S METHOD: pressure: 40-60 mmHg
- B. GOTHLIN METHOD: pressure: 35-50 mmHg

GRADE	NUMBER OF PETECHIAE	INTERPRETATION
1+	1-10	Few petechiae on the anterior part of the forearm
2+	11-20	Many petechiae on the anterior part of the forearm
3+	21-50	Multiple petechiae over the whole arm and the back of the hand
4+	>50	Confluent petechiae on the anterior part and back of the hand

2. PETECHIOMETER METHOD

- •Also known as: Suction Cup Method
- Uses **NEGATIVE PRESSURE TECHNIQUE**
- Principle: It employs the use of a modified **da Silva Melle instrument**. The cup is applied to the outer surface of the arm for a period of 1 minute at 200mmHg
- Normal value: <4 petechiae

End of presentation

- Question?
- Concerns?
- Clarifications?

- Reach me at:

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